

Comparativa de Semánticas Formales (WHILE Enriquecido)

1. Semántica Natural (NS - Big-Step)

$$\mathcal{T}_{ini} = \frac{\langle x := x + 1, \sigma_{17} \rangle \longrightarrow \sigma_{18} [\text{ass}_{\text{ns}}] \quad \langle z := z \cdot 2, \sigma_{18} \rangle \longrightarrow \sigma_{19} [\text{ass}_{\text{ns}}] \quad \langle S_1, \sigma_{19} \rangle \longrightarrow \sigma_{19} [\text{for} - \text{ff}_{\text{ns}}]}{\langle S_1, \sigma_{17} \rangle \longrightarrow \sigma_{19}} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_8 = \frac{\langle x := x + 1, \sigma_{15} \rangle \longrightarrow \sigma_{16} [\text{ass}_{\text{ns}}] \quad \langle z := z \cdot 2, \sigma_{16} \rangle \longrightarrow \sigma_{17} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_{ini}}{\langle S_1, \sigma_{15} \rangle \longrightarrow \sigma_{19}} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_7 = \frac{\langle x := x + 1, \sigma_{13} \rangle \longrightarrow \sigma_{14} [\text{ass}_{\text{ns}}] \quad \langle z := z \cdot 2, \sigma_{14} \rangle \longrightarrow \sigma_{15} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_8}{\langle S_1, \sigma_{13} \rangle \longrightarrow \sigma_{19}} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_6 = \frac{\langle x := x + 1, \sigma_{11} \rangle \longrightarrow \sigma_{12} [\text{ass}_{\text{ns}}] \quad \langle z := z \cdot 2, \sigma_{12} \rangle \longrightarrow \sigma_{13} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_7}{\langle S_1, \sigma_{11} \rangle \longrightarrow \sigma_{19}} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_5 = \frac{\langle x := x + 1, \sigma_9 \rangle \longrightarrow \sigma_{10} [\text{ass}_{\text{ns}}] \quad \langle z := z \cdot 2, \sigma_{10} \rangle \longrightarrow \sigma_{11} [\text{ass}_{\text{ns}}] \quad \mathcal{T}_6}{\langle S_1, \sigma_9 \rangle \longrightarrow \sigma_{19}} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_4 = \frac{\langle x := x + 1, \sigma_7 \rangle \longrightarrow \sigma_8 [\text{ass}_{\text{ns}}] \quad \langle z := z \cdot 2, \sigma_8 \rangle \longrightarrow \sigma_9 [\text{ass}_{\text{ns}}] \quad \mathcal{T}_5}{\langle S_1, \sigma_7 \rangle \longrightarrow \sigma_{19}} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_3 = \frac{\langle x := x + 1, \sigma_5 \rangle \longrightarrow \sigma_6 [\text{ass}_{\text{ns}}] \quad \langle z := z \cdot 2, \sigma_6 \rangle \longrightarrow \sigma_7 [\text{ass}_{\text{ns}}] \quad \mathcal{T}_4}{\langle S_1, \sigma_5 \rangle \longrightarrow \sigma_{19}} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_2 = \frac{\langle x := x + 1, \sigma_3 \rangle \longrightarrow \sigma_4 [\text{ass}_{\text{ns}}] \quad \langle z := z \cdot 2, \sigma_4 \rangle \longrightarrow \sigma_5 [\text{ass}_{\text{ns}}] \quad \mathcal{T}_3}{\langle S_1, \sigma_3 \rangle \longrightarrow \sigma_{19}} [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_1 = \frac{\langle x := 0, \sigma_1 \rangle \longrightarrow \sigma_2 \text{ [ass}_{\text{ns}}] \quad \langle z := z \cdot 2, \sigma_2 \rangle \longrightarrow \sigma_3 \text{ [ass}_{\text{ns}}] \quad \mathcal{T}_2}{\langle \textbf{for } x := 0 \textbf{ to } 8 \textbf{ do } z := z \cdot 2, \sigma_1 \rangle \longrightarrow \sigma_{19}} \text{ [for - tt}_{\text{ns}}]$$

$$\mathcal{T}_{fin} = \frac{\langle z := 1, \sigma_0 \rangle \longrightarrow \sigma_1 \text{ [ass}_{\text{ns}}] \quad \mathcal{T}_1}{\langle S_2, \sigma_0 \rangle \longrightarrow \sigma_{19}} \text{ [comp}_{\text{ns}}]$$

Sentencias Abreviadas

$$S_1 = \textbf{for } x := x + 1 \textbf{ to } 8 \textbf{ do } z := z \cdot 2$$

$$S_2 = (z := 1 ; \textbf{for } x := 0 \textbf{ to } 8 \textbf{ do } z := z \cdot 2)$$

Estados

$$\sigma_0 = \emptyset$$

$$\sigma_1 = \{z \mapsto 1\}$$

$$\sigma_2 = \{x \mapsto 0, z \mapsto 1\}$$

$$\sigma_3 = \{x \mapsto 0, z \mapsto 2\}$$

$$\sigma_4 = \{x \mapsto 1, z \mapsto 2\}$$

$$\sigma_5 = \{x \mapsto 1, z \mapsto 4\}$$

$$\sigma_6 = \{x \mapsto 2, z \mapsto 4\}$$

$$\sigma_7 = \{x \mapsto 2, z \mapsto 8\}$$

$$\sigma_8 = \{x \mapsto 3, z \mapsto 8\}$$

$$\sigma_9 = \{x \mapsto 3, z \mapsto 16\}$$

$$\sigma_{10} = \{x \mapsto 4, z \mapsto 16\}$$

$$\sigma_{11} = \{x \mapsto 4, z \mapsto 32\}$$

$$\sigma_{12} = \{x \mapsto 5, z \mapsto 32\}$$

$$\sigma_{13} = \{x \mapsto 5, z \mapsto 64\}$$

$$\sigma_{14} = \{x \mapsto 6, z \mapsto 64\}$$

$$\sigma_{15} = \{x \mapsto 6, z \mapsto 128\}$$

$$\sigma_{16} = \{x \mapsto 7, z \mapsto 128\}$$

$$\sigma_{17} = \{x \mapsto 7, z \mapsto 256\}$$

$$\sigma_{18} = \{x \mapsto 8, z \mapsto 256\}$$

$$\sigma_{19} = \{x \mapsto 8, z \mapsto 512\}$$

2. Semántica Operacional Estructurada (SOS - Small-Step)

$$\mathcal{T}_{ini} = \frac{\langle z := 1, \sigma_0 \rangle \Rightarrow \sigma_1 [\text{ass}_{\text{sos}}]}{\langle S_1, \sigma_0 \rangle \Rightarrow \langle \text{for } x := 0 \text{ to } 8 \text{ do } z := z \cdot 2, \sigma_1 \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_1 = \langle \text{for } x := 0 \text{ to } 8 \text{ do } z := z \cdot 2, \sigma_1 \rangle \Rightarrow \langle S_2, \sigma_1 \rangle [\text{for}_{\text{sos}}]$$

$$\mathcal{T}_2 = \langle S_2, \sigma_1 \rangle \Rightarrow \langle S_3, \sigma_1 \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_3 = \frac{\langle x := 0, \sigma_1 \rangle \Rightarrow \sigma_2 [\text{ass}_{\text{sos}}]}{\langle (x := 0 ; z := z \cdot 2), \sigma_1 \rangle \Rightarrow \langle z := z \cdot 2, \sigma_2 \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_4 = \frac{\mathcal{T}_3}{\langle S_3, \sigma_1 \rangle \Rightarrow \langle S_4, \sigma_2 \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\mathcal{T}_5 = \frac{\langle z := z \cdot 2, \sigma_2 \rangle \Rightarrow \sigma_3 [\text{ass}_{\text{sos}}]}{\langle S_5, \sigma_2 \rangle \Rightarrow \langle S_6, \sigma_3 \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_6 = \langle S_6, \sigma_3 \rangle \Rightarrow \langle S_7, \sigma_3 \rangle [\text{for}_{\text{sos}}]$$

$$\mathcal{T}_7 = \langle S_7, \sigma_3 \rangle \Rightarrow \langle S_8, \sigma_3 \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_8 = \frac{\langle x := x + 1, \sigma_3 \rangle \Rightarrow \sigma_4 [\text{ass}_{\text{sos}}]}{\langle (x := x + 1 ; z := z \cdot 2), \sigma_3 \rangle \Rightarrow \langle z := z \cdot 2, \sigma_4 \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_9 = \frac{\mathcal{T}_8}{\langle S_8, \sigma_3 \rangle \Rightarrow \langle S_5, \sigma_4 \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\mathcal{T}_{10} = \frac{\langle z := z \cdot 2, \sigma_4 \rangle \Rightarrow \sigma_5 [\text{ass}_{\text{sos}}]}{\langle S_5, \sigma_4 \rangle \Rightarrow \langle S_6, \sigma_5 \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{11} = \langle S_6, \sigma_5 \rangle \Rightarrow \langle S_7, \sigma_5 \rangle \text{ [for}_{\text{sos}}]$$

$$\mathcal{T}_{12} = \langle S_7, \sigma_5 \rangle \Rightarrow \langle S_8, \sigma_5 \rangle \text{ [if - tt}_{\text{sos}}]$$

$$\mathcal{T}_{13} = \frac{\langle x := x + 1, \sigma_5 \rangle \Rightarrow \sigma_6 \text{ [ass}_{\text{sos}}]}{\langle (x := x + 1 ; z := z \cdot 2), \sigma_5 \rangle \Rightarrow \langle z := z \cdot 2, \sigma_6 \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{14} = \frac{\mathcal{T}_{13}}{\langle S_8, \sigma_5 \rangle \Rightarrow \langle S_5, \sigma_6 \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\mathcal{T}_{15} = \frac{\langle z := z \cdot 2, \sigma_6 \rangle \Rightarrow \sigma_7 \text{ [ass}_{\text{sos}}]}{\langle S_5, \sigma_6 \rangle \Rightarrow \langle S_6, \sigma_7 \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{16} = \langle S_6, \sigma_7 \rangle \Rightarrow \langle S_7, \sigma_7 \rangle \text{ [for}_{\text{sos}}]$$

$$\mathcal{T}_{17} = \langle S_7, \sigma_7 \rangle \Rightarrow \langle S_9, \sigma_7 \rangle \text{ [if - tt}_{\text{sos}}]$$

$$\mathcal{T}_{18} = \frac{\langle x := x + 1, \sigma_7 \rangle \Rightarrow \sigma_8 \text{ [ass}_{\text{sos}}]}{\langle (x := x + 1 ; z := z \cdot 2), \sigma_7 \rangle \Rightarrow \langle z := z \cdot 2, \sigma_8 \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{19} = \frac{\mathcal{T}_{18}}{\langle S_8, \sigma_7 \rangle \Rightarrow \langle S_5, \sigma_8 \rangle} \text{ [comp}_{\text{sos}}^1]$$

$$\mathcal{T}_{20} = \frac{\langle z := z \cdot 2, \sigma_8 \rangle \Rightarrow \sigma_9 \text{ [ass}_{\text{sos}}]}{\langle S_5, \sigma_8 \rangle \Rightarrow \langle S_6, \sigma_9 \rangle} \text{ [comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{21} = \langle S_6, \sigma_9 \rangle \Rightarrow \langle S_7, \sigma_9 \rangle \text{ [for}_{\text{sos}}]$$

$$\mathcal{T}_{22} = \langle S_7, \sigma_9 \rangle \Rightarrow \langle S_8, \sigma_9 \rangle \text{ [if - tt}_{\text{sos}}]$$

$$\mathcal{T}_{23} = \frac{\langle x := x + 1, \sigma_9 \rangle \Rightarrow \sigma_{10} [\text{ass}_{\text{sos}}]}{\langle (x := x + 1 ; z := z \cdot 2), \sigma_9 \rangle \Rightarrow \langle z := z \cdot 2, \sigma_{10} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{24} = \frac{\mathcal{T}_{23}}{\langle S_8, \sigma_9 \rangle \Rightarrow \langle S_5, \sigma_{10} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\mathcal{T}_{25} = \frac{\langle z := z \cdot 2, \sigma_{10} \rangle \Rightarrow \sigma_{11} [\text{ass}_{\text{sos}}]}{\langle S_5, \sigma_{10} \rangle \Rightarrow \langle S_6, \sigma_{11} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{26} = \langle S_6, \sigma_{11} \rangle \Rightarrow \langle S_7, \sigma_{11} \rangle [\text{for}_{\text{sos}}]$$

$$\mathcal{T}_{27} = \langle S_7, \sigma_{11} \rangle \Rightarrow \langle S_9, \sigma_{11} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{28} = \frac{\langle x := x + 1, \sigma_{11} \rangle \Rightarrow \sigma_{12} [\text{ass}_{\text{sos}}]}{\langle (x := x + 1 ; z := z \cdot 2), \sigma_{11} \rangle \Rightarrow \langle z := z \cdot 2, \sigma_{12} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{29} = \frac{\mathcal{T}_{28}}{\langle S_8, \sigma_{11} \rangle \Rightarrow \langle S_5, \sigma_{12} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\mathcal{T}_{30} = \frac{\langle z := z \cdot 2, \sigma_{12} \rangle \Rightarrow \sigma_{13} [\text{ass}_{\text{sos}}]}{\langle S_5, \sigma_{12} \rangle \Rightarrow \langle S_6, \sigma_{13} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{31} = \langle S_6, \sigma_{13} \rangle \Rightarrow \langle S_7, \sigma_{13} \rangle [\text{for}_{\text{sos}}]$$

$$\mathcal{T}_{32} = \langle S_7, \sigma_{13} \rangle \Rightarrow \langle S_8, \sigma_{13} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{33} = \frac{\langle x := x + 1, \sigma_{13} \rangle \Rightarrow \sigma_{14} [\text{ass}_{\text{sos}}]}{\langle (x := x + 1 ; z := z \cdot 2), \sigma_{13} \rangle \Rightarrow \langle z := z \cdot 2, \sigma_{14} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{34} = \frac{\mathcal{T}_{33}}{\langle S_8, \sigma_{13} \rangle \Rightarrow \langle S_5, \sigma_{14} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\mathcal{T}_{35} = \frac{\langle z := z \cdot 2, \sigma_{14} \rangle \Rightarrow \sigma_{15} [\text{ass}_{\text{sos}}]}{\langle S_5, \sigma_{14} \rangle \Rightarrow \langle S_6, \sigma_{15} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{36} = \langle S_6, \sigma_{15} \rangle \Rightarrow \langle S_7, \sigma_{15} \rangle [\text{for}_{\text{sos}}]$$

$$\mathcal{T}_{37} = \langle S_7, \sigma_{15} \rangle \Rightarrow \langle S_8, \sigma_{15} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{38} = \frac{\langle x := x + 1, \sigma_{15} \rangle \Rightarrow \sigma_{16} [\text{ass}_{\text{sos}}]}{\langle (x := x + 1 ; z := z \cdot 2), \sigma_{15} \rangle \Rightarrow \langle z := z \cdot 2, \sigma_{16} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{39} = \frac{\mathcal{T}_{38}}{\langle S_8, \sigma_{15} \rangle \Rightarrow \langle S_5, \sigma_{16} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\mathcal{T}_{40} = \frac{\langle z := z \cdot 2, \sigma_{16} \rangle \Rightarrow \sigma_{17} [\text{ass}_{\text{sos}}]}{\langle S_5, \sigma_{16} \rangle \Rightarrow \langle S_6, \sigma_{17} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{41} = \langle S_6, \sigma_{17} \rangle \Rightarrow \langle S_7, \sigma_{17} \rangle [\text{for}_{\text{sos}}]$$

$$\mathcal{T}_{42} = \langle S_7, \sigma_{17} \rangle \Rightarrow \langle S_8, \sigma_{17} \rangle [\text{if} - \text{tt}_{\text{sos}}]$$

$$\mathcal{T}_{43} = \frac{\langle x := x + 1, \sigma_{17} \rangle \Rightarrow \sigma_{18} [\text{ass}_{\text{sos}}]}{\langle (x := x + 1 ; z := z \cdot 2), \sigma_{17} \rangle \Rightarrow \langle z := z \cdot 2, \sigma_{18} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{44} = \frac{\mathcal{T}_{43}}{\langle S_8, \sigma_{17} \rangle \Rightarrow \langle S_5, \sigma_{18} \rangle} [\text{comp}_{\text{sos}}^1]$$

$$\mathcal{T}_{45} = \frac{\langle z := z \cdot 2, \sigma_{18} \rangle \Rightarrow \sigma_{19} [\text{ass}_{\text{sos}}]}{\langle S_5, \sigma_{18} \rangle \Rightarrow \langle S_6, \sigma_{19} \rangle} [\text{comp}_{\text{sos}}^2]$$

$$\mathcal{T}_{46} = \langle S_6, \sigma_{19} \rangle \Rightarrow \langle S_7, \sigma_{19} \rangle [\text{for}_{\text{sos}}]$$

$$\mathcal{T}_{47} = \langle S_7, \sigma_{19} \rangle \Rightarrow \langle \mathbf{skip}, \sigma_{19} \rangle \text{ [if} - \mathbf{ff}_{\text{sos}}]$$

$$\mathcal{T}_{fin} = \langle \mathbf{skip}, \sigma_{19} \rangle \Rightarrow \sigma_{19} \text{ [skip}_{\text{sos}}]$$

Sentencias Abreviadas

$$S_1 = (z := 1; \mathbf{for} \ x := 0 \mathbf{to} \ 8 \mathbf{do} \ z := z \cdot 2)$$

$$S_2 = \mathbf{if} \ 0 \leq 8 \mathbf{then} \ x := 0; z := z \cdot 2; \mathbf{for} \ x := x + 1 \mathbf{to} \ 8 \mathbf{do} \ z := z \cdot 2 \mathbf{else} \ \mathbf{skip}$$

$$S_3 = (x := 0; z := z \cdot 2; \mathbf{for} \ x := x + 1 \mathbf{to} \ 8 \mathbf{do} \ z := z \cdot 2)$$

$$S_4 = (z := z \cdot 2; \mathbf{for} \ x := x + 1 \mathbf{to} \ 8 \mathbf{do} \ z := z \cdot 2)$$

$$S_5 = (z := z \cdot 2; \mathbf{for} \ x := x + 1 \mathbf{to} \ 8 \mathbf{do} \ z := z \cdot 2)$$

$$S_6 = \mathbf{for} \ x := x + 1 \mathbf{to} \ 8 \mathbf{do} \ z := z \cdot 2$$

$$S_7 = \mathbf{if} \ x + 1 \leq 8 \mathbf{then} \ x := x + 1; z := z \cdot 2; \mathbf{for} \ x := x + 1 \mathbf{to} \ 8 \mathbf{do} \ z := z \cdot 2 \mathbf{else} \ \mathbf{skip}$$

$$S_8 = (x := x + 1; z := z \cdot 2; \mathbf{for} \ x := x + 1 \mathbf{to} \ 8 \mathbf{do} \ z := z \cdot 2)$$

$$S_9 = (x := x + 1; z := z \cdot 2; \mathbf{for} \ x := x + 1 \mathbf{to} \ 8 \mathbf{do} \ z := z \cdot 2)$$

Estados

$$\sigma_0 = \emptyset$$

$$\sigma_1 = \{z \mapsto 1\}$$

$$\sigma_2 = \{x \mapsto 0, z \mapsto 1\}$$

$$\sigma_3 = \{x \mapsto 0, z \mapsto 2\}$$

$$\sigma_4 = \{x \mapsto 1, z \mapsto 2\}$$

$$\sigma_5 = \{x \mapsto 1, z \mapsto 4\}$$

$$\sigma_6 = \{x \mapsto 2, z \mapsto 4\}$$

$$\sigma_7 = \{x \mapsto 2, z \mapsto 8\}$$

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